

THE SHAPE OF A STIMULUS

Trajectory Geometry of Mouse V1 Population
Activity

30563 MATHEMATICAL MODELLING FOR NEUROSCIENCE Project

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A mouse watches three videos.

**Do its neurons code them as
three distinguishable
population states?**

Why this slice of MICrONS for this research question?



CLIP

Natural movie footage



MONET2

Grey-scale textured noise

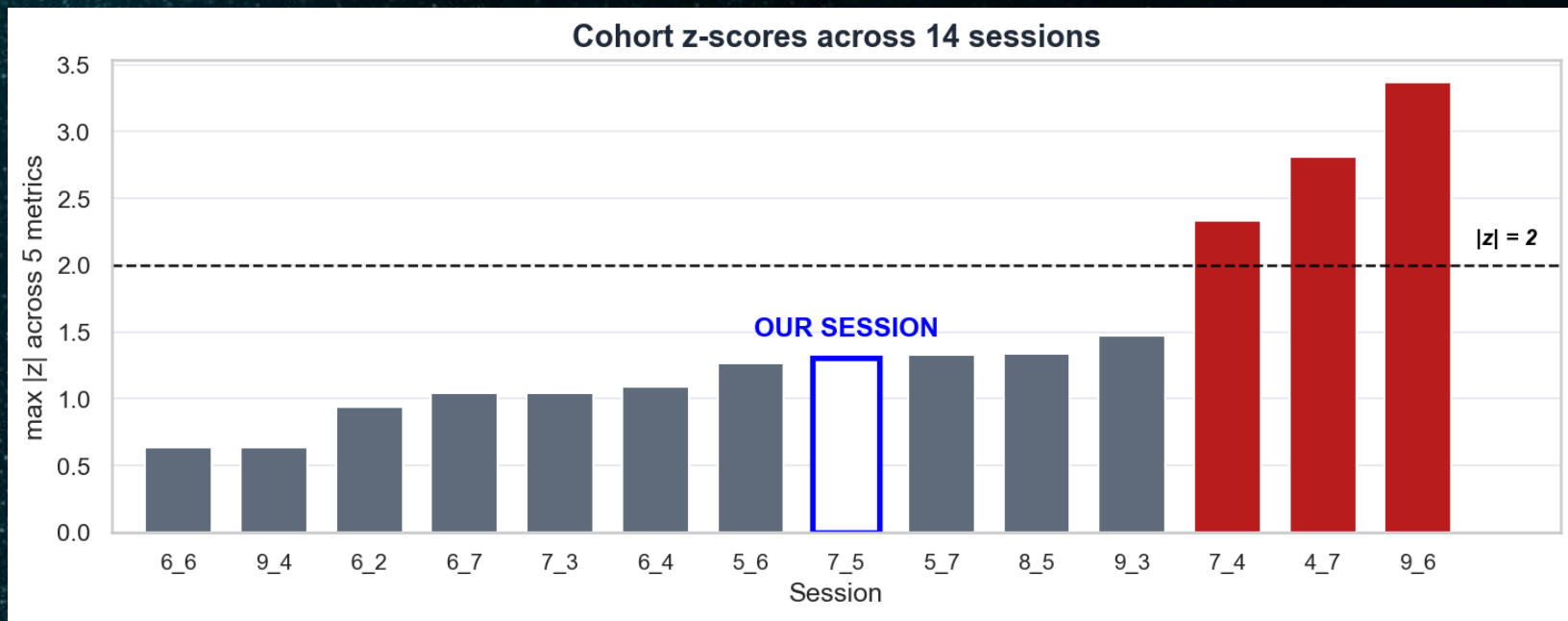


TRIPPY

Synthetic wave patterns

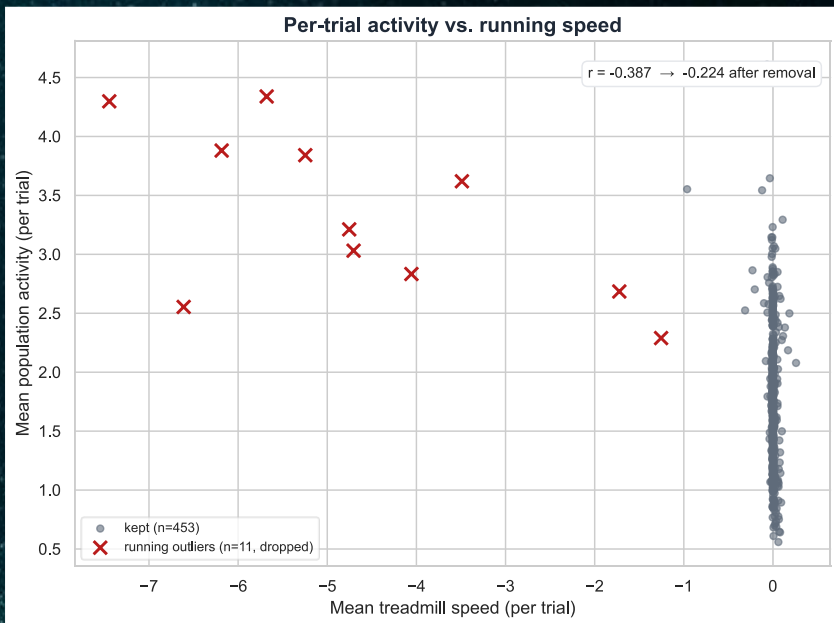
FUNCTIONAL DATA · ~8,000 NEURONS RECORDED SIMULTANEOUSLY
4 CORTICAL AREAS · V1 · LM · RL · AL

Session choice and trial structure

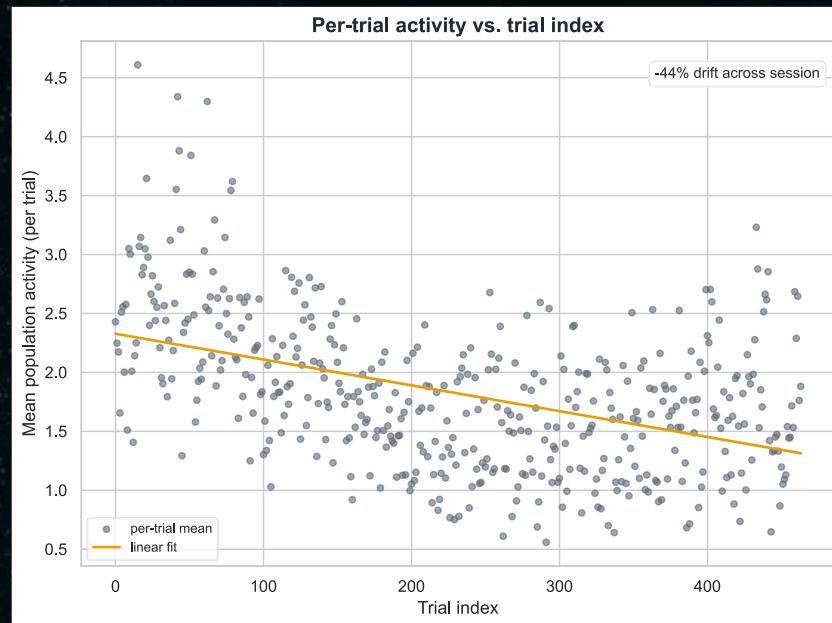


SESSION 7_5 · $|z| \leq 1.31$ ACROSS ALL METRICS · 8,194 NEURONS (COHORT MEDIAN: 8,176)
453 TRIALS · 377 CLIP · 38 MONET2 · 38 TRIPPY · INTERLEAVED

Preprocessing techniques



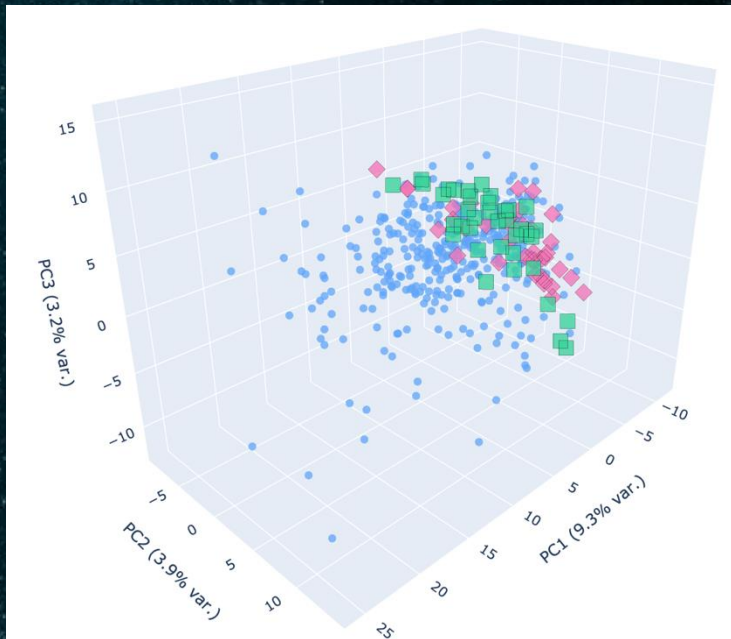
Running influences V1 — 11 outlier trials removed (2.4%)



Photobleaching — linear detrend per trial

PIPELINE: OUTLIER FILTER → LINEAR DETREND → PER-NEURON Z-SCORE

Experiment 1: trial-mean PCA



n = 453 trials · 5,485 neurons · top 3 PCs

each trial → mean activity vector → PCA

BUT

98.9%

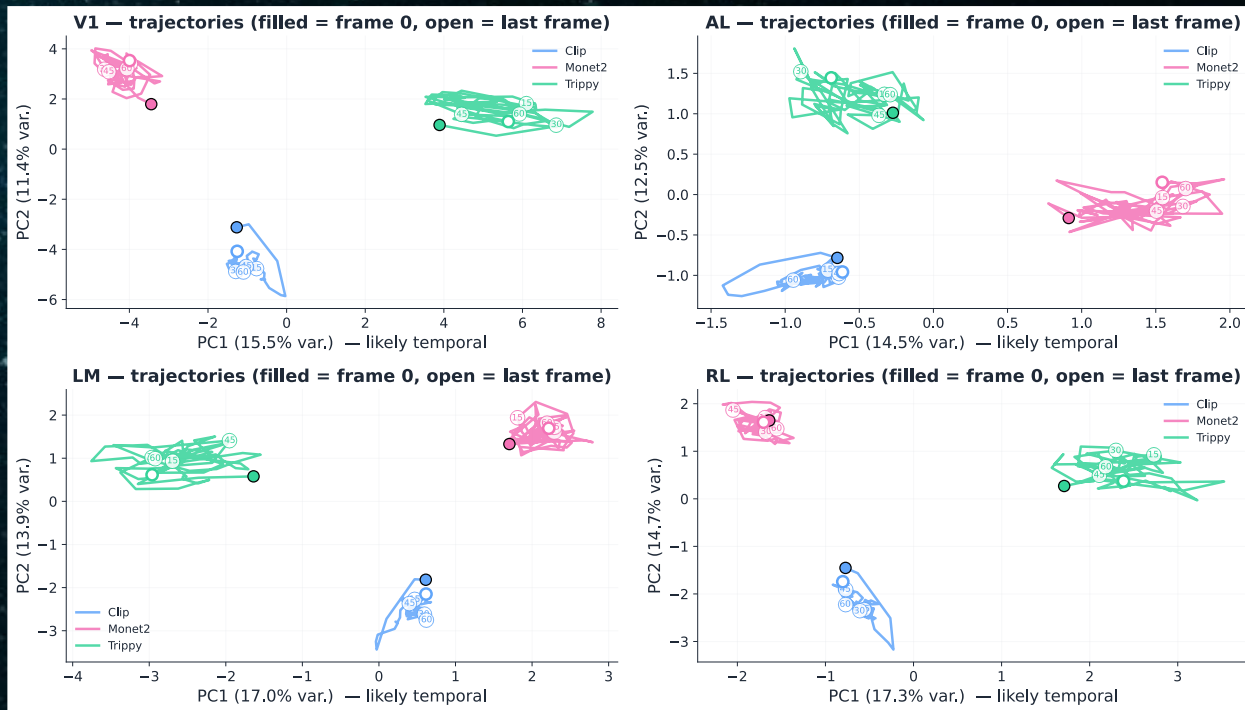
classifier accuracy on the same trial-mean vectors

majority baseline: 83.2%

LM 98.0% · RL 97.8% · AL 94.7%

THE SIGNAL IS THERE. PCA ISN'T ALIGNED WITH IT.

Experiment 2: Trial-averaged trajectories - projected into PC space



Three separated blobs in every area
BUT IS THE SEPARATION REAL?

Biased: PCA fit on real-label trial averages, then shuffled data projected through the same PCs

honest test: full feature space · no projection · all neurons

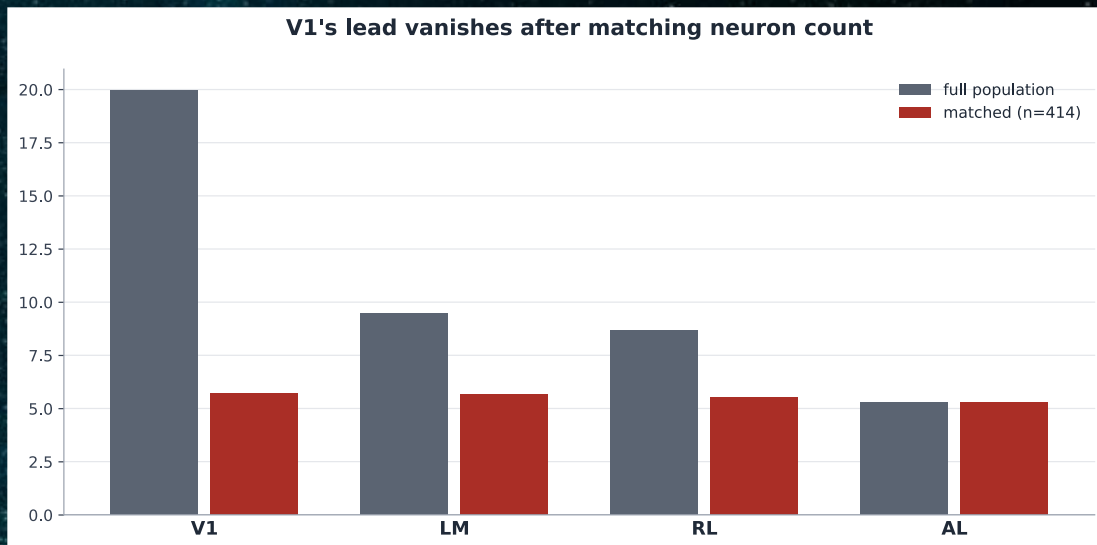


Monet2 ↔ Trippy never reliably separates in any area

THE COLLAPSE PERSISTS

Two controls—both pass

V1's apparent lead vanishes after matching neuron count.



TRIAL-COUNT CONTROL

Clip subsampled to 38 trials
(matched to Monet2 / Trippy)

DISTANCES UNCHANGED

Robust to trial-count asymmetry

Linear methods hit a ceiling
even with the time axis

Experiment 3: same method—zoomed inside Clip



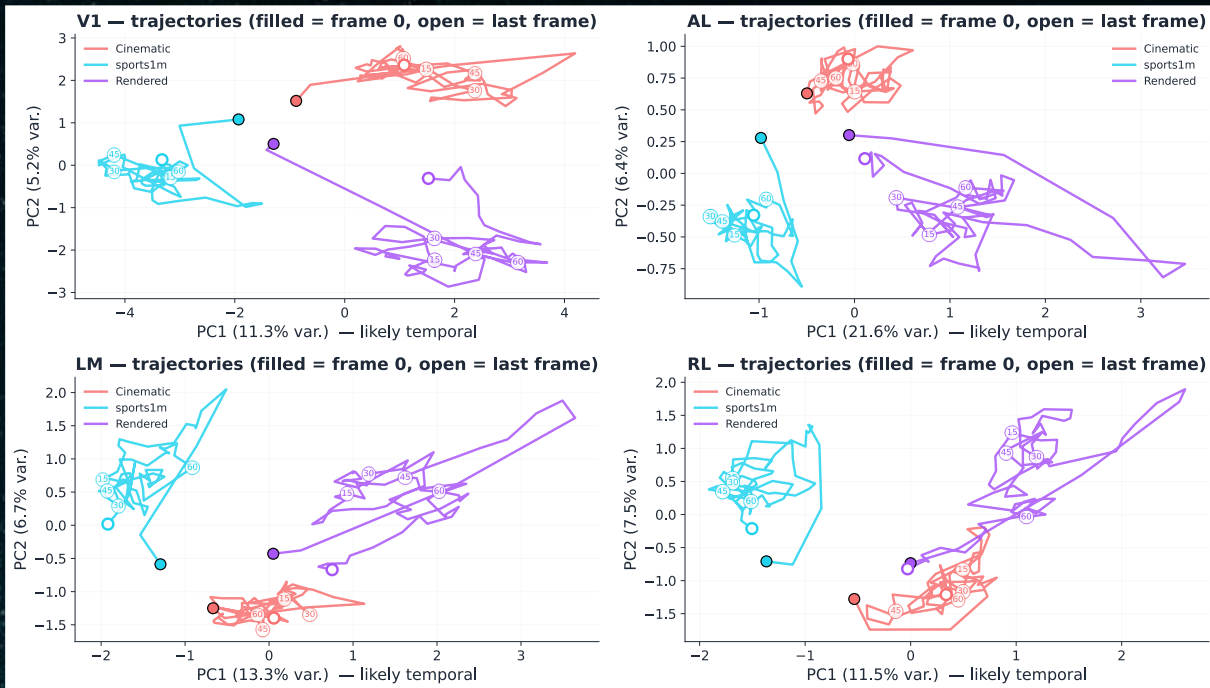
Cinematic



sport1m



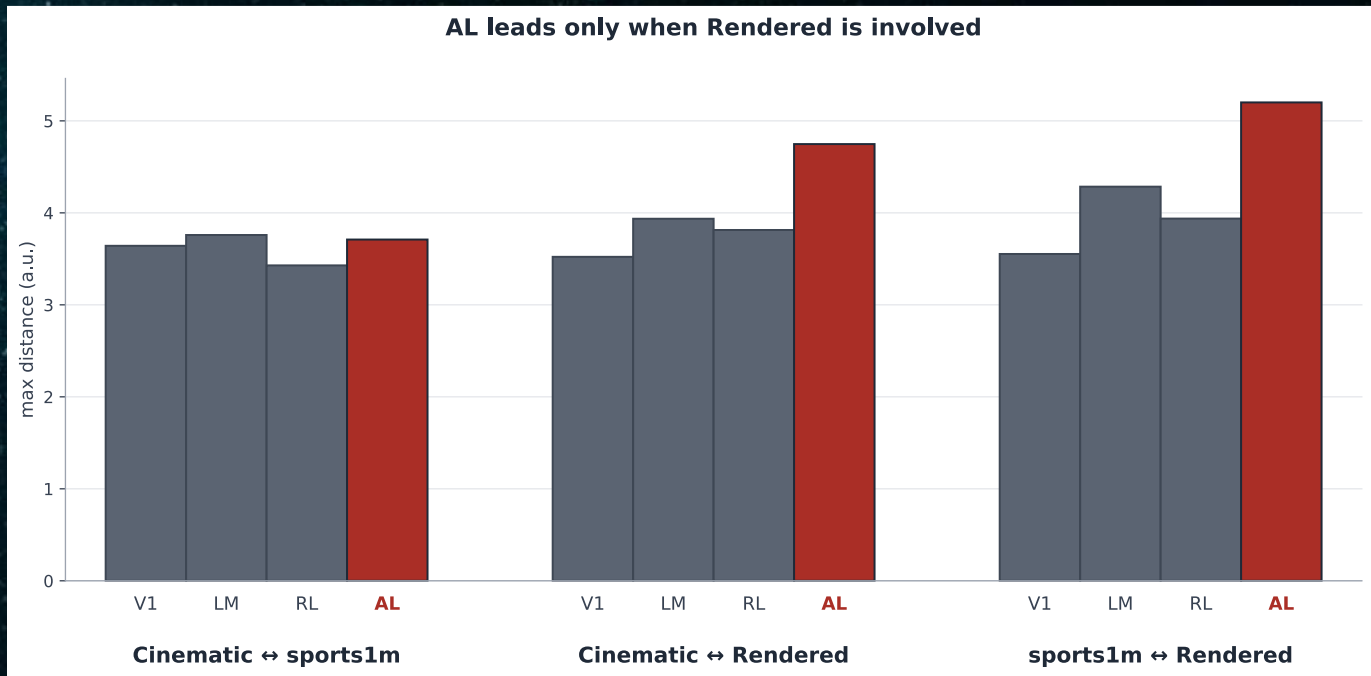
Rendered



Three sub-categories separate in every area with $p \leq 0.02$

BUT IS ONE AREA SPECIAL?

After population matching – per-neuron signal



AL is a naturalness detector, the same axis Experiment 2 found.

AL detects artificiality, not the category.

The same boundary, again.

LIMITATIONS

Linear methods may miss non-linear structure

Single session (7_5) - cohort generalization untested

Connectomics unused – structural link untested



WHAT'S NEXT

Alternative methods: **CEBRA or MANIFOLD EMBEDDINGS**
designed for exactly this regime

Cross-session replication

Functional ↔ structural link

use the connectome to test whether functional clusters reflect wiring

**Linear methods sense the naturalness boundary
but hit a ceiling for synthetic vs. synthetic**

Any questions?

THANK YOU!